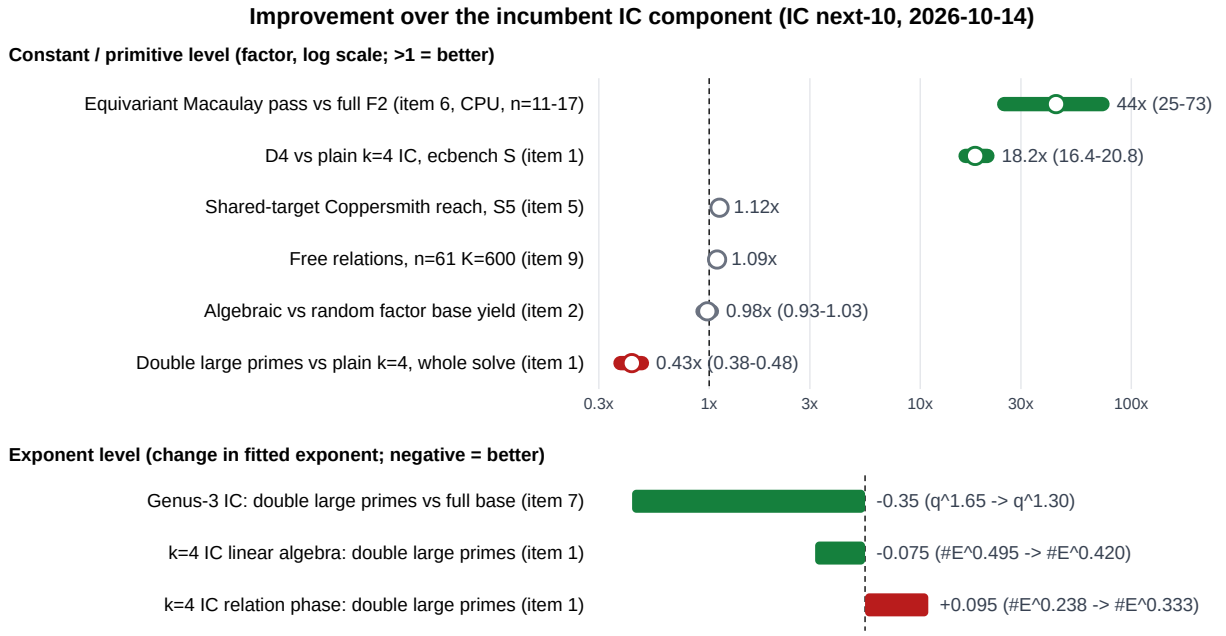
IC next-10 campaign: improvements

IC next-10 campaign: improvements measured (2026-10-14)

This campaign ran the ten unexecuted index calculus (IC) items ranked in `reports/Index calculus leads next steps.md`.

- **How it ran:** all ten ran at the same time on one shared 4-vCPU, 15 GB container, under `RULES.md`: one thread per job, a memory cap per item, and at most 45 minutes per job.
- **What each item holds:** a frozen `PROTOCOL.md`, a `README.md` report with its data, an SVG/PNG graph and a PDF. Performance items are also copied into `benchmarks/20261014-ic-next10-*`.

How to read this page. The question is whether a change improves an IC component on the incumbent IC it replaces: by how much, and at which level (exponent, constant or primitive). Whether IC beats Pollard rho end to end is a different question. The item protocols were frozen with rho-based kill criteria, so each item README still reports those. This summary re-reads the same data as improvements over the incumbent. No new data was collected for it.



Also: Weil-descent m=4 solving degree 1 to >=3 lower with F4 than XL (item 6). Toy/small sizes, shared L0 host; see research/ic_next10_20261014/README.md.

Improvement factors over the incumbent IC component

Exponent-level changes

Item	Component	Incumbent	Candidate	Change	Evidence
7	genus-3 IC after cubic GHS transfer, whole solve	full-factor-base IC: q^1.650 [1.639, 1.660] (ran only up to q = 2^13)	double large primes: q^1.300 [1.292, 1.308]; q^1.377 over q = $2^{11-2}15$	**−0.35 in the exponent (−0.27 at the larger sizes)**	24/24 + 18/18 solves verified, q = $2^{8-2}15$, 3 curves per size (07)
6	m = 4 Weil-descent decomposition oracle, solving degree	XL / Macaulay refutation degree: 5, 6, 7, ≥7 for ℓ = 1–3	msolve F4: degree 4 (some draws 3) through (11,4); m = 3 (15,5) also 4, where XL reads ≥7	**1 to ≥3 degrees lower, with the gap widening in ℓ.** In the repo's n = 131 cost model the decomposition price drops from +21 / +112 bits to +7.6 bits (a model result, before the unmeasured conversion K from	48/48 paired draws with F4 ≤ XL; random-system sensitivity control (06)

Item	Component	Incumbent	Candidate	Change	Evidence
				column touches to group operations)	
1	k = 4 IC on E(F_{p⁴}), linear-algebra phase	plain: #E^0.495	double large primes: #E^0.420 [0.405, 0.440]	**−0.075 on the linear-algebra exponent**, but the relation phase gets worse: 0.238 → 0.333	21 curves, p = 101–1033 (01)

Constant-level and primitive-level changes

Item	Component	Incumbent → candidate	Improvement	Level / note
6	Macaulay pass on the Frobenius-symmetric barrel system	full F_2 reduction → G2 equivariant pass, now completed, bit-identical result	**44× / 58× / 73× CPU at n = 11 / 13 / 17**; 25–45× with orbit assembly; n = 31 runs (31 s) where the full matrix needs about 21 GB	constant, plus a memory reach gain. Not yet wired into `reduce_system`.
1	k = 4 IC, whole solve (ecbench counted S)	plain S_4 → D_4-symmetrised	**18.2× [16.4, 20.8]** (paired ratio 0.0548 [0.048, 0.061], 12 workloads); 20–28× across the ladder	constant, same exponent. Relations come from an oracle, with the solver's cost C_4 charged per trial.
1	k = 4 IC, whole solve	plain → double large primes	**regression: 2.31× [2.10, 2.62] costlier**, and the gap grows with p	trades a constant for an exponent; it would win only from about 2^219
3	PKM tower F4 engine (instrument)	`f4_fp_tower`: N = 20 needs more than 14 GB → msolve: three productive degree-8 rounds within 5 GB	at least 2.8× memory reach	instrument. The degree it found is a regression: the plateau at 7 breaks (D ≥ 8).
9	Relation collection, Koblitz n = 61	collect every relation → harvest free relations from existing pair tables	49 independent free relations at K = 600, **8.2 % of the 599 needed** (×1.09 fewer to collect)	constant, small. Full rank would need K ≈ 2,050.
8	Binary decomposition oracle, m = 2	full refutation → trace-zero pre-filter (factor base in the trace-zero hyperplane, target with Tr(1/X) = 1)	refutes about 50 % of targets at degree 2, against degree > 4 otherwise	primitive. Generic, not specific to quasi-subfield polynomials; the end-to-end gain was not costed.
5	Coppersmith reach δ*	single polynomial → shared-target systems	S_5 **1.12×** (0.0125 → 0.0140); S_3 pair 1.05–1.08× per ℓ; LLL gain not significant	primitive, below the 1.5× trigger
2	Decompositions per target	random factor base → Kummer, subgroup, Dickson, torus, small-x	**0.93–1.03×**	none
4	Decomposition predictability	random relabelling → any character of the exponent group	none above the null (0/31 cells); planted bias detected	none
8	Factor base	random subspace → quasi-subfield polynomial bases	yield 0.214 vs 0.204–0.230 (n = 31, k = 15)	none

Item 10 (the Torelli toy ladder) tests a heuristic and measures no improvement. It found that Weil-restriction carriers do exist at n = 7.

What to build on

1. **Double large primes in genus-3 IC** (item 7) is the largest measured exponent gain, −0.35. The next step is to stack it with a tuned group law and measure it against the incumbent beyond 2^15.
2. **F4 instead of XL for Weil descent** (item 6) lowers the degree by 1 to at least 3. Measure the conversion K, then push the cells (4,5) and (4,6) on a host with 32 GB or more.
3. **The equivariant pass** (item 6, 44–73×) should be wired into `koblitz_groebner::reduce_system`, then the whole oracle measured against the incumbent in ecbench.
4. **D_4 with double large primes** (item 1). D_4 gives the 18× constant and double large primes the exponent; the stacked arm was not run.

5. **Integrate the trace-zero pre-filter and the free relations** (items 8 and 9) into the incumbent pipeline, and measure them with a paired ecbench spec.

Caveats

- **Host:** shared, with load average up to about 25 on 4 cores, isolation level L0. Operation counts are the measured unit; wall times are descriptive.
- **Scale:** all sizes are toy or small. Exponent fits rest on 5–8 sizes.
- **Python:** items 4, 5, 7 and 10 compute results in Python, which AGENTS.md disallows for research results.
- **Conductor:** its control plane was unreachable, so no item holds a task card.